

Q1-Read the questions carefully and choose one correct answer for each question:

(15 marks)

1. If $D = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 7 & 2 \end{bmatrix}$, the value of element d_{22} in the matrix $[D (D^T)]$ is:
 a) 11 b) 29 **(c) 25** d) 53
2. If the function $f(x) = e^x - 4$ has a root within the interval $x_0 = 1$ and $x_1 = 2$, what is the next value of the approximation of (x_{r+1}) after (x_r) using the bisection method.
 a) 0.745 b) 1.75 **(c) 1.25** d) 1.44
3. When a set of computed results are in close agreement with the true value, but do not agree with each other, then this set of computed results is described as:
 a) inaccurate and imprecise **(b) accurate and imprecise**
 c) inaccurate and precise d) accurate and precise
4. Matrix which does not have an inverse is classified as:
(a) singular matrix b) non-singular matrix c) linear matrix d) unidentified matrix
5. The equation $f(x)$ is given as $x^3 + 4x + 1 = 0$. Considering the initial approximation at $x_0 = 1$ then the approximated value of x_{r+1} , applying Newton Raphson method to find the root is:
 a) 0.15886 b) 0.11286 c) 0.12936 **(d) 0.14286**
6. If $\begin{vmatrix} a & -b & s \\ k & t & r \\ m & g & u \end{vmatrix} = 5$, then the value of $\begin{vmatrix} 3a & -b & s \\ 3k & t & r \\ 3m & g & u \end{vmatrix}$ is:
 a) 5 **(b) 15** c) 30 d) 135
7. If A and B are 5x5 matrices with $|A| = 7$ and $|B| = 2$, then the value of $|A^T B|$ is:
 a) 3.5 **(b) 14** c) 7 d) 0.285
8. To find the root of the equation $\cos 2x - 5x$ in the Fixed-Point iteration method by assuming $x_0 = 0.2$, the first approximated value of x_1 will be:
 a) 0.24996 b) -0.07894 c) 0.19999 **(d) 0.18421**
9. The False Position method converts the function algebraically into the form $g(x) = x$, then convergence of the iterations occurs if:
 a) $|g'(x)| = 1$ b) $|g'(x)| > 1$ **(c) $|g'(x)| < 1$** d) $|g'(x)| > 2$
10. The determinant of $A = \begin{bmatrix} 3 & 5 & 2 \\ 0 & 4 & 5 \\ 0 & 0 & 3 \end{bmatrix}$ is:
(a) 36 b) -69 c) 20 d) 60

Q2- Perform two iterations to find the root for the following equation in which the real root of the equation lies in the interval (0.8, 0.9): (8 marks)

$$\sin x - x^2 = 0$$

Note: Apply the Secant method, show all your calculations.

Q3- Use the Gauss Elimination method to solve the following system of equations: show all your calculations in details: determine the values of x, y, z: (7 marks)

$$x + y + z = 2$$

$$2x + 3y + z = 11$$

$$5x - 2y + z = -6$$

End of Questions

Q2 $x_{i-1} = 0.8$ $f(0.8) = 0.0773560909$
 $x_i = 0.9$ $f(0.9) = -0.02667309$

① $x_{i+1} = 0.8 - \frac{f(0.8) \times (0.9 - 0.8)}{-0.02667309 - 0.0773560909} = 0.87435992$

2) $x_{i-1} = 0.9$ $f(0.9) = -0.02667309$
 $x_i = 0.87435992$ $f(0.87435992) = +$
 $x_{i+1} = 0.87435992 - \frac{f(0.87435992) \times (0.9 - 0.87435992)}{f(0.9) - f(0.87435992)}$
 $x_{i+1} = 0.8766593974$
 $E\% = \frac{0.8766593974 - 0.87435992}{0.8766593974} \times 100\% = 0.26\%$

Q3 $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 5 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 11 \\ -6 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & -7 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ -16 \end{bmatrix}$

① $-8z = 33$

$z = -4.125$

② $y - z = 7 = y - (-4.125) = 7$
 $y = 2.875$

③ $x + 2.875 - 4.125 = 2$
 $x = 3.25$